

Spark Optimization Techniques

Boosting Performance in Big Data Workflows



Optimize Join Strategies

- ✓ **Broadcast Join:** Use when one table is small.
- ✓ **Shuffle Hash Join:** For medium tables, hash both sides.
- ✓ **Sort-Merge Join:** Ideal for large, sorted datasets.



Enable Adaptive Query Execution (AQE)

- ✓ Dynamically adjusts join types.
- ✓ Coalesces small partitions to **reduce shuffle**.
- ✓ Optimizes skewed data at runtime.



Use Caching Efficiently

- ✓ Cache **DataFrames** only when reused often.
- ✓ Use in-memory storage wisely.
- ✓ Avoid caching large, temporary data.



Reduce Shuffle & Spill

- ✓ **Repartition** to optimize partitions.
- ✓ **Filter early** to reduce data movement.
- ✓ Tune `spark.sql.shuffle.partitions` setting.



Key Performance Tips:

- ✓ **Tune Executor Memory & Cores**
- ✓ **Use Optimal File Formats** (Parquet/Delta)
- ✓ **Monitor** with the *Spark UI & Logs*